

Trained



train



Large labelled training set

Corpus-specialized features

Standard



Cross-env.



Training-free, prior work



Frozen patch features

Standard

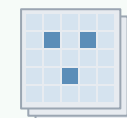


Cross-env.



Codebook
or
Segmentation CLS attention
Prior-selected / aggregated features

Ours



Frozen patch features

Standard



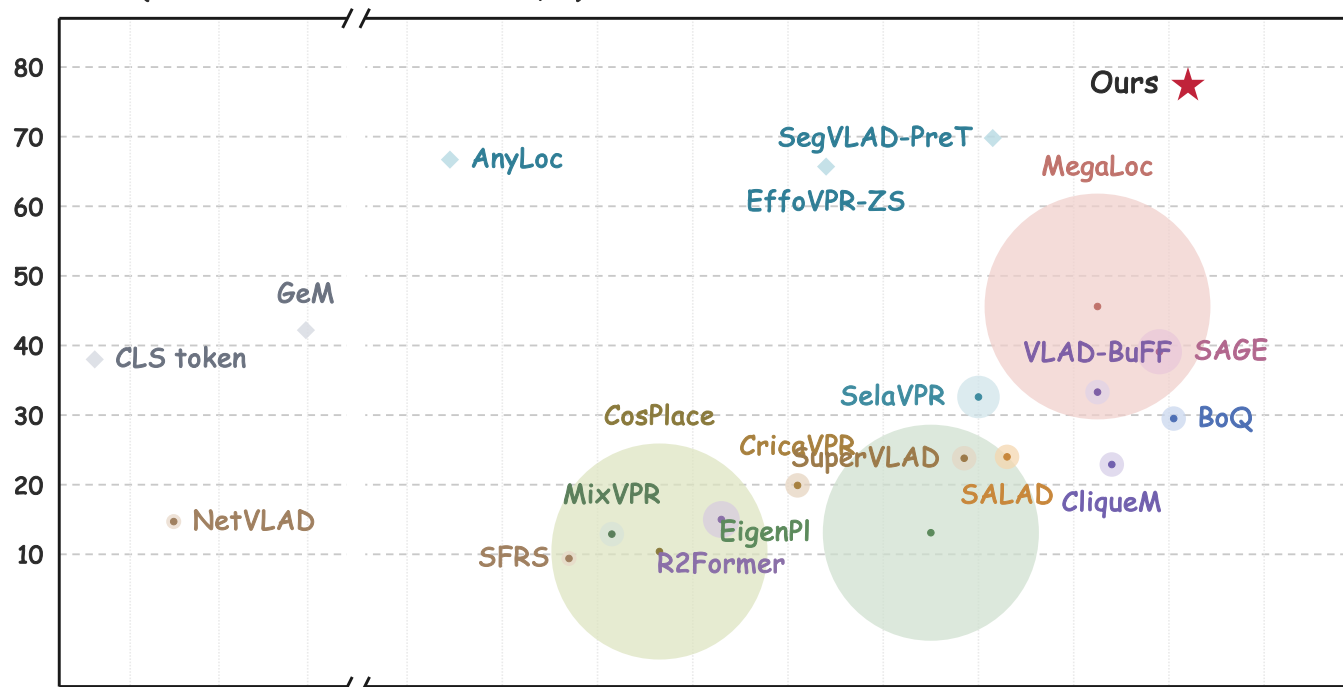
Cross-env.



Within-image
Balanced, matchable patch evidence
Across-image

Supervision cost vs. cross-environment recall

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

★ ours (training-free)

♦ training-free

● trained

● 1 million images

● 5 million images

Recall across 17 benchmarks

